

with absolutely continuous distribution function, we refer to the classical book of Shao [26]. In extension of the celebrated results of Hoeffding [21], the weak limit of U-statistics has been derived under several weak dependence measures, see the contributions of Eagleson [19], Denker [13], Carlstein [7], Babbal [3], Dehling and Wendler [12] and Leucht [23].

Our contribution in this direction is a functional CLT for U-statistics for so-called L^p - m -approximable processes, i.e., processes $X = (X_t)_t$ such that

- (i) each X_t admits a Bernoulli shift representation $X_t = f(\varepsilon_t, \varepsilon_{t-1}, \dots)$, where the map $f: S^\infty \rightarrow M$ is some measurable function and $(\varepsilon_t)_t$ are i.i.d. random elements with values in some measure space S ;
- (ii) the m -dependent approximation process $(X_t^{(m)})_t$ which is given by

$$X_t^{(m)} = f(\varepsilon_t, \dots, \varepsilon_{t-m+1}, \varepsilon_{t-m}^{(t)}, \varepsilon_{t-m-1}^{(t)}, \dots), \quad t \in \mathbb{Z},$$

satisfies

$$\sum_{m=1}^{\infty} \mathbb{E}[d(X_0, X_0^{(m)})^p]^{1/p} < \infty. \quad (2.2)$$

Here, for each fixed $t \in \mathbb{Z}$, the sequence $(\varepsilon_j^{(t)})_j$ is an independent copy of the original innovation process $(\varepsilon_j)_j$.

In order to derive the weak (functional) limit of (2.1) we follow Hoeffding's approach and consider the decomposition of the U-statistic in a main and a degenerate part:

$$U_n(h) = \theta + \frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} h_2(X_i, X_j) + \frac{1}{n} \sum_{i=1}^n h_1(X_i) + \frac{1}{n} \sum_{j=1}^n h_1(X_j),$$

where

$$h_1: M \rightarrow \mathbb{R}, \quad x \mapsto \mathbb{E}[h(x, X')] - \theta \quad \text{with } X' \sim \mathbb{P}_X$$

and

$$h_2: M \times M \rightarrow \mathbb{R}, \quad (x, y) \mapsto h(x, y) - \theta - h_1(x) - h_1(y).$$

Define for $0 \leq s, t \leq 1$ and $1 \leq i \leq n$ (for $n \geq 2$)

$$\begin{aligned} a_n(s, t) &:= \frac{1}{n(n-1)} \sum_{i=1}^{\lfloor ns \rfloor} \sum_{j=1}^{\lfloor nt \rfloor} \mathbb{1}\{i \neq j\} \\ &= \frac{1}{n(n-1)} \max\{(\lfloor nt \rfloor - 1)\lfloor ns \rfloor, (\lfloor ns \rfloor - 1)\lfloor nt \rfloor, 0\} \end{aligned}$$

and

$$b_{n,i}(t) := \frac{1}{n-1} \sum_{j=1}^{\lfloor nt \rfloor} \mathbb{1}\{i \neq j\}$$

$$= \frac{1}{n-1} \max \left\{ (\lfloor nt \rfloor - 1) \mathbb{1}\{i \leq \lfloor nt \rfloor\} + \lfloor nt \rfloor \mathbb{1}\{i > \lfloor nt \rfloor\}, 0 \right\}.$$

The corresponding two-parameter partial sum process $U_n(h): [0, 1]^2 \rightarrow \mathbb{R}$ is given by

$$U_n(h)(s, t) := \frac{1}{n(n-1)} \sum_{i=1}^{\lfloor ns \rfloor} \sum_{\substack{j=1 \\ i \neq j}}^{\lfloor nt \rfloor} h(X_i, X_j) \quad (2.3)$$

$$\begin{aligned} &= a_n(s, t)\theta + \frac{1}{n(n-1)} \sum_{i=1}^{\lfloor ns \rfloor} \sum_{\substack{j=1 \\ i \neq j}}^{\lfloor nt \rfloor} h_2(X_i, X_j) \\ &\quad + \frac{1}{n} \sum_{i=1}^{\lfloor ns \rfloor} h_1(X_i) b_{n,i}(t) + \frac{1}{n} \sum_{j=1}^{\lfloor nt \rfloor} h_1(X_j) b_{n,j}(s). \end{aligned} \quad (2.4)$$

The symmetric function h_2 yields the so-called degenerate part of the U -statistic, that is

$$\mathbb{E}[h_2(x, X_1)] = \mathbb{E}[h(x, X_1)] - \theta - \mathbb{E}[h(x, X')] + \theta = 0 \quad \forall x \in M,$$

because $\mathbb{E}[h_1(X_1)] = \theta - \theta = 0$. Hence, we expect the functions h_1 to contribute most to $U_n(h)$.

In order to derive our FCLT, we need to strengthen the assumptions in two ways (we make this rigorous in Assumption 2.1 below): On the one hand, additionally to the approximation property (2.2), we assume a specific decay of the sequence $(\mathbb{E}[d(X_0, X_0^{(m)})])_m$. On the other hand, we require some control on the regularity and growth of the kernel h to insure that the degenerate part in (2.4) asymptotically vanishes.

Assumption 2.1 ($(\mathbf{p}, \mathbf{q}, \varrho)$). Let $p, q \geq 1$ be Hölder conjugates of each other, i.e., $1/p + 1/q = 1$, and let $\varrho \in (0, 1]$.

(A1) The kernel h is ϱ -Hölder continuous w.r.t. the product metric on $M \times M$, i.e.,

$$|h(x, y) - h(x', y')| \leq L d_{M \times M}((x, y), (x', y'))^\varrho = L (d(x, x') + d(y, y'))^\varrho \quad (2.5)$$

for some $L > 0$. Moreover, the kernel satisfies the moment condition

$$\sup_{U, V \sim \mathbb{P}_X} \mathbb{E}[|h(U, V)|^q]^{1/q} =: K(q) < \infty, \quad (2.6)$$

where the supremum is taken over all couplings (U, V) with each marginal distribution equal to $\mathbb{P}_X$.

(A2) Let $X = (X_t)_t$ be an M -valued L^p - m -approximable process, whose Fréchet variance exists, i.e., we have $\mathbb{E}[d(X_t, x)^2] < \infty$ for some $x \in M$. Moreover, X is approximable in the sense that

$$\sum_{m=1}^{\infty} m \mathbb{E}[d(X_0, X_0^{(m)})^{\varrho p}]^{1/p} = \sum_{k=1}^{\infty} \sum_{m \geq k} \mathbb{E}[d(X_0, X_0^{(m)})^{\varrho p}]^{1/p} < \infty. \quad (2.7)$$

Remark 2.2. Assumption $(\mathbf{p}, \mathbf{q}, \varrho)$ entails the Hölder continuity of h_1 and h_2 , moreover, the approximation property at a linear rate can be guaranteed under a stronger polynomial rate. In detail we have the following.

- *Approximation property.* If $\varrho = 1$, the kernel is even Lipschitz continuous and we can neglect ϱ in (2.7): the process X is L^p - m -approximable at a linear rate m .

If, however, $0 < \varrho < 1$, condition (2.7) holds provided that

$$\sum_{m=1}^{\infty} m^{(1+\sigma)/\varrho} \mathbb{E}[d(X_0, X_0^{(m)})^p]^{1/p} < \infty \quad (2.8)$$

for some $\sigma > 1 - \varrho$, i.e., the process X is L^p - m -approximable at a polynomial rate $m^{(1+\sigma)/\varrho}$. Indeed, applying Jensen's and Hölder's inequality in this latter case shows

$$\begin{aligned} & \sum_{m=1}^{\infty} m \mathbb{E}[d(X_0, X_0^{(m)})^{\varrho p}]^{1/p} \\ & \leq \sum_{m=1}^{\infty} \left(m^{1+\sigma} \mathbb{E}[d(X_0, X_0^{(m)})^p]^{\varrho/p} \right) m^{-\sigma} \\ & \leq \left(\sum_{m=1}^{\infty} m^{(1+\sigma)/\varrho} \mathbb{E}[d(X_0, X_0^{(m)})^p]^{1/p} \right)^{\varrho} \left(\sum_{m=1}^{\infty} m^{-\sigma/(1-\varrho)} \right)^{1-\varrho}. \end{aligned}$$

- *Hölder continuity.* h_1 inherits the ϱ -Hölder continuity of the kernel h with the same Hölder constant L because

$$\begin{aligned} |h_1(x) - h_1(y)| & \leq \mathbb{E}[|h(x, X') - h(y, X')|] \\ & \leq L \mathbb{E}[\|(d(x, y), d(X', X'))\|_r^{\varrho}] \leq L d(x, y)^{\varrho}. \end{aligned} \quad (2.9)$$

Moreover, the function h_2 is ϱ -Hölder continuous as well:

$$\begin{aligned} & |h_2(x, y) - h_2(x', y')| \\ & \leq |h(x, y) - h(x', y')| + |h_1(x) - h_1(x')| + |h_1(y) - h_1(y')| \\ & \leq L \left(\|(d(x, x'), d(y, y'))\|_r^{\varrho} + d(x, x')^{\varrho} + d(y, y')^{\varrho} \right) \\ & \leq L \left((d(x, x') + d(y, y'))^{\varrho} + d(x, x')^{\varrho} + d(y, y')^{\varrho} \right) \\ & \leq 3L \left(d(x, x') + d(y, y') \right)^{\varrho}, \end{aligned} \quad (2.10)$$

where the last inequality follows from the triangle inequality.

Theorem 2.3 (FCLT for $U_n(h)$). *Granted Assumption 2.1 (p, q, ϱ) is satisfied for some $p \geq 2$. Then*

$$\left(\sqrt{n}(U_n(h)(s, t) - st\theta) \mid 0 \leq s, t \leq 1 \right) \Rightarrow Y$$